Smart Contract Security & Vulnerability Analysis

Autonomous Comprehensive Synthesis • Authored by Cognexa Autonomous AI

Subject:	Smart Contract Security & Vulnerability Analysis	**Date:**	Current Operational Cycle
Engine:	Cognexa Core 2.0 (LangGraph + Vector Memory)	**Security:**	Verified & Sandboxed

1. Executive Summary

Smart contracts on Ethereum are immutable once deployed. A single vulnerability can lead to permanent financial loss.

1. Checks-Effects-Interactions (CEI) Pattern

Always validate preconditions using require statements, update internal contract state before transferring external assets.

- Validate balance >= withdrawal amount
- Deduct balance before calling external recipient
- Prevents reentrancy state manipulation

2. Reentrancy Vulnerability Analysis

Handing control to external untrusted addresses before zeroing balances allows recursive invocations.

- Exploited in The DAO hack (2016)
- Solution: OpenZeppelin ReentrancyGuard nonReentrant modifier

3. Pre-Deployment Security Audits

Automated static analysis with Slither and fuzzing with Echidna ensures contracts undergo formal verification.